1. Clinical Question & Core Hypothesis

Clinical Question

Intravesical BCG provokes local immune activation and often induces lymphoid organization, yet many urothelial carcinomas with apparent immune infiltration/TLS fail to achieve durable responses—why does organized local immunity sometimes not translate into BCG sensitivity? 2) Tier 1 paths used: the path linking PI3-kinase pathway activation → PD-L1 (CD274) and the path describing BCG → upregulation of PD-L1 provide a mechanistic backbone: tumor-intrinsic PI3K activity can induce PD-L1, and BCG itself can further elevate PD-L1, both converging to suppress antitumor T cell function. I therefore focused on the PI3K→CD274 axis. 3) Tier 2 supporting evidence: nodes such as TMPRSS2/ERG and PSMA (stem-cell–like programs / lineage features) suggest tumor-intrinsic oncogenic contexts that may co-select for immune-evasive phenotypes and impact durability of response; these reinforce but do not define the core mechanism. 4) Endpoint choice rationale: therapy_response (BCG response) was chosen because the path centers on BCG-induced immune biology and PD-L1–mediated suppression; this endpoint is the most clinically actionable for hypothesis-driven intervention (combination therapy). 5) Deliberate exclusions: I excluded prostate-specific edges (e.g., androgen deprivation and TMPRSS2/ERG as a core anchor) from the primary claim because they are less directly relevant to intravesical BCG in urothelial carcinoma; PSMA and TMPRSS2/ERG are included as secondary_modifiers for exploratory stratification. I also treated the semantically odd edge suggesting PD-L1 upregulates BCG as an annotation artifact and instead used the biologically plausible relationships (BCG induces PD-L1; PI3K induces PD-L1)?

Working Hypothesis

Tumor-intrinsic PI3K signaling activation via PIK3CA drives upregulation of PD-L1 (CD274) on urothelial carcinoma cells, dampening TLS-like lymphoid aggregate–associated T cell activity during intravesical BCG. Clinically, TLS-like aggregates in tumors with high PIK3CA activity and CD274 expression will show reduced response to BCG, but concurrent PI3K or PD-L1 blockade with BCG is predicted to enhance TLS-driven anti-tumor immunity and improve outcomes.

Element	Value
Imaging feature	TLS-like lymphoid aggregates
Anchor gene(s)	CD274, PIK3CA
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

CD274: CD274) and the path describing BCG → upregulation of PD-L1 provide a mechanistic backbone: tumor-intrinsic PI3K activity can induce PD-L1, and BCG itself can further elevate PD-L1, both converging to suppress antitumor T cell function.

PIK3CA: Selected as a core mechanistic driver from Tier 1 knowledge graph analysis.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core

Secondary	TMPRSS2/ERG	Supporting modifier from knowledge graph
Secondary	PSMA	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: The interaction between the imaging feature and anchor gene predicts clinical outcome.

Prediction 2: Spatial analysis will confirm the mechanistic link at the tissue level.

Prediction 3: Functional perturbation will establish causality.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Novel mechanistic hypothesis derived from knowledge graph analysis.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- TMPRSS2/ERG: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- PSMA: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.